COURSE OUTLINE----Fall 2022

CSI 5760 /ECE 5730, Embedded system Design with FPGA

ECE CRN: 41476. CSI CRN: 42314

Instructor: Professor. Subra Ganesan
Office: 440 EC, Phone: (248)370-2206; Email: ganesan@oakland.edu

Lecture: Mondays and Wednesdays **7:30 to 9:17 PM** Room number: WH 102 Wilson Hall 102

- **We do meet in person for class lectures.**
- **Labs are done by you at OU in the Lab room EC 562. There may not be a Lab TA.**
- **Labs are cooperative environment. Students can help each other. Do not copy the programs fully. Initially form a team of two to do the lab exercises.**
- **Fall semester starts on September 1, 2022. Classes end on December 5.**
- First day of class for ECE 5730 is September 7.
- **Last day of class for ECE 5730 is December 5- Final exam is later.**

Laboratory: There are no assigned lab sessions for this course. You are expected to do the lab work at OU or home and submit lab reports as they are due**.**

Office Hours: Any time by prior Appointment and also after the lecture.

TEXT Books: (Suggested) Need not busy any text book

By JIEN-CHUNG LO, "Modern Digital Designs with EDA. VHDL and FPGA" ISBN 978-986-91529-0-7. Terasic Inc., https://www.terasic.com.tw/cgi-bin/page/archive.pl?Language=English&CategoryNo=&No=945 (Suggested Text book) – **If you do not get the book, it is ok**.

Reference books

Pong P. Chu, Cleveland State University, "**EMBEDDED SOPC DESIGN WITH NIOS II PROCESSOR AND VHDL EXAMPLES- pdf copy of the book will be available on the Moodle. 723 pages.** ePDF ISBN: 978-1-118-14650-7, **A JOHN WILEY & SONS, INC., PUBLICATION**

Software: We use Quartus Prime Lite edition, Version 21.1.1 for windows. It occupies 14 GB space in your laptop. You can down load it free from https://www.intel.com/content/www/us/en/software-kit/736572/intel-quartus-prime-lite-edition-design-software-version-21-1-1-for-windows.html
We will use DE1 SOC boards and they will be available in the Lab room or the boards will be given on loan to you. You need to return them at the end of the semester.

Optional Material:

If interested buy one board from Intel.Intel/Altera Board, DE1-SoC. https://www.terasic.com.tw/cgi-bin/page/archive.pl?Language=English&No=836 Cost $ 175 for academic + shipping

Use MOODLE to upload homework, lab assignments, exam answers etc.

PREREQUISITES: CSE378, CSE505, 506, CSE 508 or equivalent courses on Digital logic, design, Assembly or C programming.

Holidays: October 21, Friday 2022 to October 23, Sunday, 2022(both inclusive) are holidays. November 24 to November 27 (both inclusive) are holidays for Thanksgiving.

Exams: Exam 1 in the class room on October 5 and Exam 2 is on November 16 Only 2 Exams. No final exam.
Catalog COURSE DESCRIPTION:

Topics covered include the use of hardware description languages such as VHDL/Verilog and C in the design of embedded systems containing an FPGA, CPU design, device drivers for FPGA cores, high-level design tools to specify, simulate and synthesize designs to FPGAs, and design examples. Hardware and software design; project-oriented course.

Topics Covered details:

1) Elements of Embedded Systems Design
2) Real Time systems Requirements
3) Field Programmable devices
4) VHDL based design and design of Logic modules
5) Simple CPU design
6) Tools for design and Prototyping
7) Device Drivers for FPGA cores
8) Design with embedded processors
9) Design of FPGA based embedded systems
10) Low Power FPGA based Systems
11) Automotive applications

COURSE objective:
Upon successful completion of the course, students will be able to:

- Write VHDL and C language codes to implement Digital system on FPGA devices.
- Use an A/D converter (Codec) to read analog signals into FPGA board
- Generate pulse-width-modulation (PWM) signals on a FPGA board for controlling the speed of a DC motor or the position of a servo
- Describe how to design using FPGA with microcontroller core.
- Work in a team environment to design a microprocessor based system and communicate the results in a written report and an oral presentation.

At periodic intervals, we will check whether the course objectives have been met.

Presenatation:

Each student has to make one presentation on one of the topics during October. Topic will be assigned by random draw. You submit the presentation as PPT with voice on Moodle. This presentation is different from Project Presentation. This carries 5% of the total grade. Hence do your best in this activity.

LABORATORY

There will be lab assignments using FPGA development boards, with emphasis on **design**. The laboratory projects will be both hardware and software assignments. Students should work as a team or individually in the labs. You can work on the lab assignments at any time using the FPGA boards. **Each student** has to

Submit on or before **November 30, 2022,** all the lab Programs listings at the moodle. For each lab include **objective, a flow chart, hardware interface details, and VHDL/C programs** with comments.

PROJECT

Each student (team) must work on a term project during the second half of the course. The project must be completed and tested before the due date mentioned below.

Project title and brief abstract is due on: November 13
Progress report I (Hardware/ software detail and component list) due on: November **20**
Presentation in the lecture room December 5
Demonstration of the project in the laboratory: December ~~5~~
Final Project report is due on: December 6, 2022. - Submit on Moodle

New Grading Scale

Oakland University switched from a numeric to a letter grade as follows.

			Excellent
95 -100	A	4.0	
90 - 94	A-	3.7	
85 - 89	B+	3.3	Good Graduate Students need B or higher
80 - 84	B	3.0	
75 - 79	B-	2.7	
72 - 74	C+	2.3	Satisfactory
68 -71	C	2.0	
62 - 67	C-	1.7	
57 - 61	D+	1.3	Minimal Pass
50- 56	D	1.0	
<50	F	0.0	Fail

EXAMINATION AND GRADE POLICY.

Dates may change if there are reasons like weather or so.

Exam I: (Closed book) ***October 5 Wednesday*** at 7:30 pm in-class **20%**
Exam II: (Open book;) ***Nov 16 wednesday*** at 7: 30 pm in-class **20%**
Laboratory Assignments **20%**
Project: Report/Demonstration10+10 **20%**
Presentation made in October **5%**
Home work …… 15%

Important Note: A minimum of 50% grade in each of Exam 1, Exam 2, Presentation, Project, Lab assignments and Homework are needed to get a passing grade. If you do not get 50% in any one of these then you will get a Fail grade. No exception.

GRADING POLICY: Grading policy may be modified if the average of the exams/score is very low. One should try to get 50% or above in each of the above, i.e. Exams, Project, Lab, HW.
Course Evaluation: Students do the on-line course evaluation at the end of the semester.
Academic Conduct: Students are expected to practice and uphold standards of academic integrity and honesty. Examples of dishonesty: cheating in exams, labs, and home work; Plagiarizing the work of others, unauthorized collaborations on computer assignments. Please refer the Graduate or Undergraduate catalog for details.

LAB REPORT

Note: Even though you do the lab as a team, the lab report is **submitted by each student**. Use the report submission guideline sheet enclosed with the first day handout.
First describe briefly about the lab setup and board. For each lab include objective, a flow chart, hardware interface details, and code with comments. Labs emphasize design of software, hardware and interface. Include in the submission, details or data sheets of A/D, D/A or any other special chip you have used in the lab. There is no need to include the FPGA chip data sheet.
Board
We use the following board in the lab. DE1- SOC sold by Terasic, product of Intel (previously Altera).
https://www.terasic.com.tw/cgi-bin/page/archive.pl?Language=English&No=836
Cost $ 175 for academic + shipping. You need not buy the board. We will give one board for your use. You need to return the board at the end of the semester to get a grade



Contents are

- DE1-SoC Board
- DE1-SoC Quick Start Guide
- Type A to B USB Cable
- Type A to Mini-B USB Cable
- Power DC Adapter (12V)

We use: **Quartus Prime Lite Edition version 18.1(for 32 bit) or 20.1 (for 64 bit)** software for development. It is free for students. Need Windows Ten OS, or Linux/Ubuntu
https://www.intel.com/content/www/us/en/software/programmable/quartus-prime/download.html

The DE1-SoC Development Kit presents a robust hardware design platform built around the Altera System-on-Chip (SoC) FPGA, which combines the latest dual-core Cortex-A9 embedded cores with industry-leading programmable logic for ultimate design flexibility. Users can now leverage the power of tremendous re-configurability paired with a high-performance, low-power processor system. Altera's SoC integrates an ARM-based hard processor system (HPS) consisting of processor, peripherals and memory interfaces tied seamlessly with the FPGA fabric using a high-bandwidth interconnect backbone. The DE1-SoC development board includes hardware such as high-speed DDR3 memory, video and audio capabilities, Ethernet networking, and much more.
The DE1-SOC Development Kit contains all components needed to use the board in conjunction with a computer that runs the Microsoft Windows XP or later (*64-bit OS and Quartus II 64-bit are required to compile projects for DE1-SoC*).